7. Short introduction to Scala

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Concurrent programming (CC3040) 2025/2026

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https://fm-dcc.github.io/cp2526

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CISTER - Research Centre in
Real-Time & Embedded
Computing Systems

Module 2: programming with Java's memory model

Blocks of sequential code running concurrently and sharing memory:

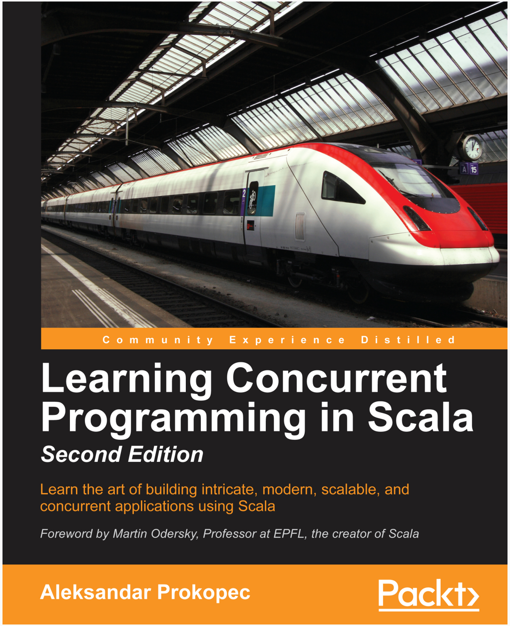
- What is **Scala** and why using it?
- Concurrency in Java and its memory model
- Basic concurrency blocks and libraries
- Futures and promises
- Actor model

We will be **less formal**

- focus on concepts and programs
- study operators and libraries
- tool support with **Scala**

We will have **hands-on**

- Practical programming exercises
- Apply the concepts we learn


Community Experience Distilled
Learning Concurrent Programming in Scala
Second Edition
Learn the art of building intricate, modern, scalable, and concurrent applications using Scala
Foreword by Martin Odersky, Professor at EPFL, the creator of Scala
Aleksandar Prokopec
Packt

Motivation

Why Scala?

Used by many modern concurrency frameworks

- Syntactic flexibility
- Programming models as Embedded Domain Specific Languages
- Many useful features

Safe language

- Automatic garbage collection
- Automatic bound checks
- No pointer arithmetic
- Static type safety

Java interoperability

- Compiled to Java bytecode
- Can use existing Java libraries
- Good interaction with Java's rich ecosystem
- Chosen by some Java-compatible frameworks

Executing Scala

Running with SBT

```scala
object SquareOf5 extends App {
  def square(x: Int): Int = x * x
  val s = square(5)
  println(s"Result: $s")
}
```

Call stack vs. object heap

where are values stored?

Concurrent threads:

do not share the call stack,

share the object heap,

Running with SBT

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Call stack vs. object heap

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Concurrent threads:

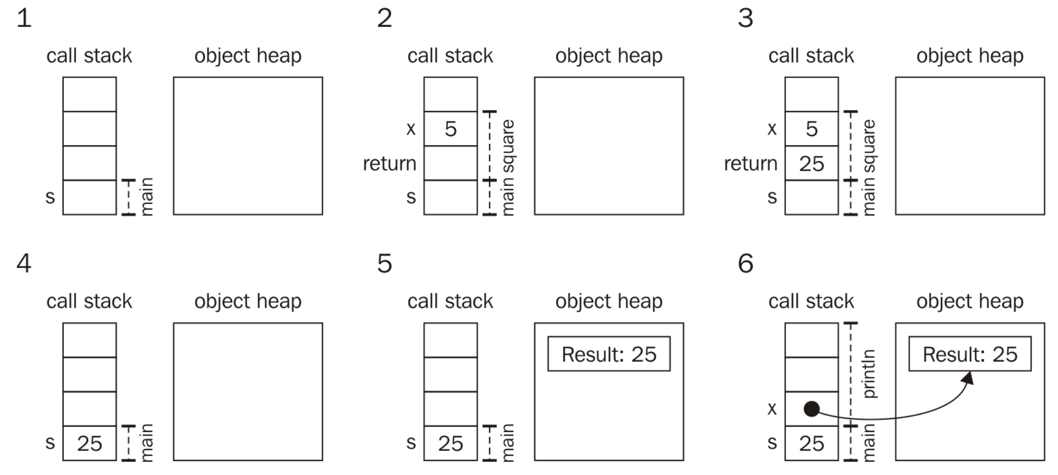
do not share the call stack,

share the object heap,

Local: SBT (practical lessons)

Online: `https://scastie.scala-lang.org/auN6B4mVSiyfE0s7RZ37xQ`

Stack and Heap



[in *"Learning Concurrent Programming in Scala"*, pg. 19]

Scala in a nutshell

Creating a Printer class

```scala
class Printer(val greeting: String) {
  def printMessage(): Unit =
      println(greeting + "!")
  def printNumber(x: Int): Unit = {
    println("Number:␣" + x)
  }
}
```

One `greeting` parameter

Two methods

```scala
class Printer(val greeting: String) {
  def printMessage(): Unit =
      println(greeting + "!")
  def printNumber(x: Int): Unit = {
    println("Number:␣" + x)
  }
}
```

Using the class

```scala
val printy = new Printer("Hi")
               // instantiate
printy.printMessage() // ?
printy.printNumber(5) // ?
```

Singleton objects

```
object Test {
  val Pi = 3.14
}
```

Using the object

```
val x = Test.Pi * 5 * 5
    // no need to instantiate
```

Traits (similar to Java Interfaces)

```scala
trait Logging {
  def log(s: String): Unit // just declared
  def warn(s: String) = log("WARN:␣" + s)
  def error(s: String) = log("ERROR:␣" + s)
}
class PrintLogging extends Logging {
  def log(s: String) = println(s)
}
```

Using traits

```scala
val x = new PrintLogging
val y = new Logging {
  def log(s:String): Unit =
    println(s)
}
```

Type parameters

```scala
class Pair[A,B](val fst: A, val snd: B)
```

Using Pair

```scala
val x: Pair[Int,String] =
    new Pair(4,"a")
val y = new Pair(2,5) //
    infer type
```

Lambdas (anonymous functions)

```
val twice_a: Int=>Int = (x:Int) => x*2
val twice_b = (x:Int) => x*2
val twice_c: Int=>Int = x => x*2
val twice_d: Int=>Int = _ * 2
```

Using lambdas

```
val x = twice_a(4)
```

Byname parameters (lazy)

```
def runTwice(body: =>Unit) = {
  body
  body
}
```

Using Byname

```
runTwice { // prints "Hello" twice
  println("Hello")
}
```

“for” expressions and comprehension

```scala
for (i <- 0 until 10) println(i)
// equivalent to
0.until(10).foreach(i => println(i))

val negatives_a =
  for (i <- 0 until 10) yield -i
val negatives_b =
  (0 until 10).map(i => -1 * i)
```

"for" expressions and comprehension

```scala
for (i <- 0 until 10) println(i)
// equivalent to
0.until(10).foreach(i => println(i))

val negatives_a =
  for (i <- 0 until 10) yield -i
val negatives_b =
  (0 until 10).map(i => -1 * i)
```

```scala
val pairs_a =
  for (x <- 0 until 4;
       y <- 0 until 4) yield (x, y)
val pairs_b =
  (0 until 4).flatMap(x =>
    (0 until 4).map(y =>
      (x, y)))
```

Scala collections and string interpolation

Common collections:

`Seq[T]`, `List[T]`, `Set[T]`, `Map[K,V]`

```
val msgs_a: Seq[String] =
    Seq("Hello", "world!")
val msgs_b: List[String]=
    "Hello"::"World"::Nil
val msgs_c: Set[String]=
    Set("Hello", "world!")
val msgs_d: Map[String,Int]=
    Map("Hello"->5, "world!"->6)
```

String interpolation:

```
val number = 7
val msg =
  s"After␣$number␣comes␣${number+1}!"
```

Pattern matching

```scala
val successors =
  Map(1 -> 2, 2 -> 3, 3 -> 4)
successors.get(5) match {
  case Some(n) =>
    println(s"Successor is: $n")
  case None    =>
    println("Could not find successor.")
}
```

```scala
trait IntOrError
case class MyInt(i:Int)
    extends IntOrError
case class MyError(e:String)
    extends IntOrError

//...
def show(ie: IntOrError) {
  ie match {
    case MyInt(i)   =>
        println(s"Number: $i")
    case MyError(e) =>
        println(s"Error: $e")
  }
}
```

Pattern matching (in Scala 3)

```scala
val successors =
  Map(1 -> 2, 2 -> 3, 3 -> 4)
successors.get(5) match
  case Some(n) =>
    println(s"Successor␣is:␣$n")
  case None    =>
    println("Could␣not␣find␣successor.")
```

```scala
enum IntOrError:
  case MyInt(i:Int)
  case MyError(e:String)

//...
def show(ie: IntOrError) =
  ie match
    case MyInt(i)   =>
        println(s"Number:␣$i")
    case MyError(e) =>
        println(s"Error:␣$e")
```

Try-catch

```scala
trait IntOrError
case class MyInt(i:Int)       extends
    IntOrError
case class MyError(e:String) extends
    IntOrError

//...
def getInt(ie: IntOrError) {
  ie match {
    case MyInt(i)    => i
    //case MyError(e) => ???
    // (match error if an error is found)
  }
}
```

```scala
def showInt(ie:IntOrError) =
  try {
    println(s"Int␣value:␣$
        {getInt(ie)}")
  } catch {
    case _: Throwable =>
        println("Got␣some␣
        error␣-␣probably␣not␣
        an␣int.")
  } finally {
    // always executes
    // (for side-effects, not
        to return results)
    println("Done␣showing")
  }
```

Operator overloading

```scala
class Position(val x: Int, val y: Int) {
  def +(that: Position) =
    new Position(x + that.x, y + that.y)
  def *(n: Int) =
    new Position(x * n, y * n)
}
```

Using operators

```scala
val p1 = new Position(3,4)
val p2 = p1 + p1 * 2 //?
```

Package objects

File `src/main/scala/cp/lablessons/package.scala`

```
package cp

package object lablessons {
  def log(msg: String): Unit =
    println(
      s"${Thread.currentThread.getName}:␣$msg"
    )
}
```

The `log` function is used throughout these lessons

Requires starting with

`package cp.lablessons`

Wrap up

- Stack and Heap
- Singleton objects
- Traits (similar to Java Interfaces)
- Type parameters
- Lambdas (anonymous functions)
- Byname parameters (lazy)
- "`for`" expressions
- "`for`" comprehension
- Scala collections and string interpolation
- Pattern matching
- Try-catch
- Operator overloading
- Package objects